

Design and Build Lab — Syllabus

School Year: 2026–2027

Instructor: Matt Bombich

Email: mbombich@vicksburgschools.org

Room: 70

Office Hours: Monday–Thursday, 2:45–3:15 pm

Prerequisites: None — open to all students.

College Credit: Pending KVCC articulation agreement

Course Description

Design and Build Lab is a full-year, project-based course where you use professional design and fabrication tools to make real things. Projects range from precision CAD-modeled parts to laser-cut signs, silicone molds, and student-designed automated systems. You will use the Engineering Design Process, Autodesk Fusion CAD software, 3D printers, CNC machines, a laser cutter, and basic electronics — guided by a portfolio that tracks your work all year.

Every project is designed to build a specific set of skills, and skills compound across the year. By the end, you will have a portfolio of finished work that demonstrates what you can design, make, and solve.

Course Objectives

By the end of the year, you will be able to:

- Apply the Engineering Design Process to plan and execute fabrication projects.
 - Model 3D parts and assemblies in Autodesk Fusion 360.
 - Fabricate parts using 3D printers, a CNC router, a laser cutter, and hand tools.
 - Solder basic electronic circuits and understand fundamental circuit concepts.
 - Work in teams to deliver finished, functional products on a timeline.
 - Reflect on design and fabrication decisions in a digital portfolio.
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Materials & Tools

All materials are provided for class projects. Personal project materials may require student purchase — details given per project.

Category	Items
Software	Autodesk Fusion (CAD), Google Workspace
Fabrication	Bambu and Creality 3D printers (PLA, PETG), CNC router, laser cutter/engraver, plasma CNC, drill press, hand tools, calipers
Electronics	Soldering irons, LEDs, resistors, power supplies, Arduino (P5 only)
Materials	Acrylic, MDF, polycarbonate, silicone casting compound, misc. hardware
Safety	Safety glasses, gloves, hearing protection (provided)
Resources	Class portfolio site, class hub

Grading

Category	Weight
Portfolio Deliverables	50%
Quizzes	20%
Midterm & Final Exam	20%
Safety, 5S & Participation	10%

Portfolio Deliverables (50%) are the core of your grade. Project milestones — design documents, fabrication logs, photos of finished work, and reflections — are submitted through your digital portfolio. Each deliverable has a rubric visible before you submit. Point values reflect the complexity of the work.

Quizzes (20%) are given at the end of major skill units to assess understanding of content and skills covered.

Midterm & Final Exam (20%) are cumulative assessments given at the end of each semester.

Safety, 5S & Participation (10%) — You begin each semester with 50 points. Ten points are deducted per incident, including: not wearing required PPE, leaving tools or materials out after use, failing to participate in 5S cleanup, or other unsafe behaviors. Serious or repeated violations are referred to administration regardless of remaining points.

Project Overview

Projects are completed roughly in sequence. Earlier projects build skills used in later ones.

Project	Title	Core Skills
P1	Fidget Spinner	CAD modeling, 3D printing, iterative fit/finish, individual
P2	LED Sign / Wall Art	CNC/plasma cutting, basic circuits, soldering, team project
P3	Acrylic Ornament	Laser cutting/engraving, CNC, finishing, intro circuits, individual
P4	Silicone Molds	CAD mold design, 3D printing, casting and demolding, individual
P5	Automated Plant Watering	Arduino, sensors, relay control, enclosure design, team project
P6	Capstone	Student-driven; all skills; team project

Pacing for each project is posted on the [Class Hub](#).

Student Leadership

FIRST Robotics is available to students at all levels as an opportunity to apply engineering skills in a competitive team environment and develop teamwork, time management, and professional communication. The Capstone Project includes a formal presentation of your engineering work to a real audience.

Classroom Policies

Safety — Non-Negotiable

PPE is required whenever shop tools or electronics are in use. A safety violation results in a 10-point deduction from

your Safety, 5S & Participation grade and immediate removal from the work area. Repeated violations will be referred to administration.

Attendance

This class is hands-on — you cannot make up fabrication time from home. Excused absences receive full make-up opportunities.

Late Work

Portfolio deliverables are accepted up to 3 school days late with a 10% per day penalty. After 3 days, work is accepted for partial credit at the teacher's discretion. Talk to Mr. Bombich before the deadline if you need an extension.

Academic Integrity

AI tools may be used to understand concepts or generate design ideas — but submitted work must reflect your own thinking and decisions. Copying is handled per school policy.

Accommodations

Accommodation plans are fully supported. Please discuss your needs privately and early — accommodations work best when we plan ahead together.

Contact & Resources

Portfolio	mbombich-robotics.github.io/CTE
Class Website	mbombich-robotics.github.io/CTE/hub.html — all assignments, rubrics, pacing, and announcements
Email	mbombich@vicksburgschools.org — replies within 24 hours on school days
Tutorial	Tuesdays & Thursdays — request your assignment by end of day Monday or Wednesday. Students with missing work are assigned automatically.